

Metro Interstate Traffic Volume — Data Pipeline

Methodology & Findings Report

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1. Overview

This report summarises the Python data pipeline built to clean, validate, and engineer features from the Metro Interstate Traffic Volume dataset — hourly records of holiday flags, temperature, precipitation, cloud cover, weather description, and traffic volume. The pipeline is implemented as a sequence of functions (cleaning, outlier handling, feature engineering, visualisation, and a command-line query application), each instrumented with structured logging so that every change made to the raw data is auditable end to end.

2. Methodology

Data cleaning. Text fields (e.g. holiday) were standardised in case and whitespace without corrupting genuine missing values; timestamps were parsed and un-parseable rows dropped; both exact duplicate rows and duplicate timestamps (a known artefact of overlapping weather-station reports in this dataset) were removed. Temperature was converted from Kelvin to Celsius before any Celsius-based validation was applied.

Outlier and impossible-value handling. A distinction was drawn between physically impossible values and genuine statistical outliers. Impossible readings — temperatures below -50°C (the dataset's known 0 K sensor-error pattern) and rainfall above 200 mm/hour (versus a real-world hourly record of roughly 305 mm) — were imputed using the monthly median, falling back to the dataset-wide median if an entire month was affected.

Feature engineering. Four categories of ML-ready features were derived from the cleaned data:

- Time — hour, day of week, a weekend indicator, and cyclical sine/cosine encodings of hour and day of week, so that adjacent time points (e.g. hour 23 and hour 0) remain numerically close.
- Weather — one-hot encoding of the low-cardinality weather_main category, plus derived indicators (precipitation present, total precipitation, overcast, clear, severe weather) built from weather_description and cloud cover.
- Numerical scaling — z-score and min-max scaled versions of temperature and traffic volume, computed directly with NumPy/Pandas.
- Target variable — a three-class congestion_category (Low / Medium / High), derived from the 33rd/66th percentiles of traffic volume rather than fixed or mean-centred thresholds.

Logging and the query application. Every stage logs at INFO level for routine confirmations and WARNING level whenever data was modified, with counts, via Python's logging module writing to both console and a persistent pipeline.log file. A command-line application (app.py) was then built on top of the engineered features, exposing three query commands — query-traffic, high-traffic, and weather-traffic — via arg parse subcommands.

3. Key Findings

Data quality. The raw dataset contained several known, well-documented data-quality issues consistent with sensor or logging faults rather than genuine readings: 0 K temperature entries (which convert to an impossible -273.15°C), rainfall values in the thousands of millimetres per hour, and duplicate timestamps from overlapping weather-station reports. Handling these explicitly, rather than silently averaging over them, was necessary; left uncorrected, they would have distorted downstream statistics such as mean temperature and any temperature-based correlation with traffic volume.

Distribution shape drove the target design. Traffic volume was found to be mildly left-skewed (mean 3,259.82 vs. median 3,380.00) with high relative variability (standard deviation $\approx 61\%$ of the mean). This ruled out a symmetric, mean $\pm$ std threshold for the congestion target, since it would not split the data evenly under that skew; a quantile-based (tertile) split was used instead, guaranteeing three roughly

balanced classes regardless of distribution shape — an important property for a balanced classification target in any later modelling step.

Outliers were preserved, not discarded. Statistical outliers in traffic volume were capped rather than removed, on the basis that unusually high or low traffic readings are plausibly real congestion or free-flow events rather than measurement error distinct in kind from the physically impossible sensor readings, which were corrected outright.

4. Conclusion

The pipeline produces a cleaned, feature-engineered dataset with a documented, auditable trail of every correction made, and a working command-line tool for querying it.